

Experimental study on contamination of building exhaust emissions in urban environment under changes of stack locations and atmospheric stability



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ABSTRACT

Contamination on exhaust emissions from short stacks on a building is the major concern in the urban environment. Dispersion of exhaust emissions around the building was investigated experimentally in a boundary layer wind-tunnel experiments under changes of stack locations and atmospheric stability conditions. Experiments with different conditions of atmospheric stability were conducted using an isolated cubical building model. Tracer gas (C_2H_4) was released from different stack locations on the building roof. Flow, temperature and dispersion fields were analyzed and measured using a split fiber probe, a cold-wire thermometer, and high-response flame ionization detector. Atmospheric diffusion was examined at different distances downwind of the leeward side of the cubical building. Flow and dispersion characteristics studied included mean velocity, turbulent kinetic energy, mean and fluctuation concentrations around the building. The results show that the stable condition increased the vertical velocity around the building. The unstable condition increased the longitudinal velocity around the building. The concentration distributions increased for the right stack and decreased for the left stack on the building. There is an increase in the concentrations with stable conditions, and a decrease in the concentrations with unstable conditions.

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1. Introduction

Contamination of toxic and odorous gases emitted from short stacks on a building in urban environment are critical for protecting the air quality due to potential health hazards associated with them. Furthermore, gaseous exhaust emissions from such stacks around buildings can affect indoor air quality by entering the emitting building or an adjacent building in the near vicinity. Many parameters effect of gaseous emissions, such as atmospheric stability, stack height and locations, wind velocity and direction, the height and geometry of the building and exhaust momentum. Regarding the stack locations and atmospheric stability, it is known that the changes of stack locations and stability conditions are most important in the pollutant dispersion around buildings. The near-ground atmospheric layer typically exhibits a stable stratification during nighttime and unstable stratification during daytime. For moderate to higher wind speeds, however, sufficient atmospheric turbulence is generated so that near-neutral stability prevails.

There have been several experimental studies conducted to investigate various influences on pollutant dispersion, as it is transported from the stack to intakes in urban area. Li and Meroney [1] carried out wind tunnel experiments on a building model to assess effluent concentration in the near-wake region. The results showed that effluent concentrations were reduced by the presence of a sharp edge on the building. The majority of previous research on rooftop stack emissions has been done for buildings (e.g., White and Stein [2]; Schulman and Scire [3]; Murakami et al. [4]; Thompson [5]; Wilson et al. [6]). These studies have shown that an increase in stack height or exhaust speed decreases roof-level concentrations. Furthermore, Lowery and Jacko [7] studied the effects of building parapets and varying intake configurations on the dilution of a pollutant between a rooftop stack and building intake. The parapets were found to produce a cavity zone that extends above the roof by as much as twice the physical height of the parapet; increasing stack height had little effect on dispersion until the stack extended beyond this cavity region.

Recently, Saathoff et al. [8] evaluated the effect of rooftop structures (RTSs) on the dispersion of building exhaust, using wind tunnel experiments with an isolated building for four different full-scale equivalent heights, from 15 to 70 m. The influence of the RTS on plume behavior for normal wind diminished as building height

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increased, because of an increase in size of the separated flow region above the roof. Caciolo et al. [9] investigated the wind impact on stack effect in the openings experimentally, i.e. the single-sided natural ventilation. They observed the increasing of the outflow air rate through the windward opening under the wind. The reduced air rate was noticed in the case of leeward opening.

Hajra et al. [10] examined the effects of near-field pollutant dispersion of upstream buildings in the built environment and compared them to the ASHRAE 2007 model. The effect of spacing (S) between the buildings and stack from the upwind edge of the emitting building (X) were also investigated. Data show that within the recirculation zone, a change in the longwind dimension of the upstream building has a negligible effect on the dilution of emissions from the downwind building. Plume geometry is largely governed by the upwind dimensions of the upstream building. Gładyszewska-Fiedoruk and Gajewski [11] studied the effects of wind on the stack ventilation performance in an average kindergarten in Białystok (Central Europe). The measurements were taken when indoor and outdoor temperatures were similar so as to minimize the differences in stack effect, i.e. reducing the influence of temperature difference on the ventilation rate. Wind has a considerable influence on the air rate in the stack ventilation system; the research proved this rate could change up to 350%.

On the other hand, the effect of near-field pollutant dispersion in the case of downstream buildings in the urban environment was studied in a wind tunnel by Hajra and Stathopoulos [12]. The results showed that the height and acrosswind dimension of the downstream building, as well as spacing between the buildings, are critical parameters for assessing plume dilution. The effects of wind direction, exhaust speed, stack location, stack height, and RTS crosswind width on severity of the downwash effect of an RTS, representing a typical RTS effect on plume dispersion, was assessed by Gupta et al. [13]. Downwash produced by the RTS significantly increased roof-level concentration, depending on building height, stack location, stack height, exhaust speed, wind direction and RTS crosswind width.

The effect of atmospheric stability on pollutant dispersion in an urban environment has been the subject of numerous investigations. However, there are limited experimental studies of the effect of atmospheric stability on dispersion within the wake of a building. Yang and Meroney [14], Kothari et al. [15], Higson et al. [16] and Mavroidis et al. [17] studied dispersion in the building wake under different atmospheric stratifications. Uehara et al. [18] examined the effects of thermal stratification on flow in and above urban street canyons in a wind tunnel. Yassin et al. [19] experimentally investigated pollutant dispersion over a built-up area in the field, and in a wind tunnel under different thermal stabilities and wind directions. The results indicate that thermal stability and wind direction are important parameters in the study of atmospheric diffusion. Santos et al. [20,21] examined during field experiment the effects of averaging time on building-influenced atmospheric dispersion, under different atmospheric stabilities. Very few experimental studies have dealt with the effect of atmospheric stability on dispersion.

Despite, there are very few of these studies suggest the potential impact of different stack locations and stability conditions around buildings, and there is no work treating the effect of stack locations and atmospheric stratification on dispersion patterns of emissions. Consequently, there is still a need for further study of the effect of different stack locations and atmospheric stability on dispersion. The present study aims to investigate, using wind tunnel experiments, mean velocity, turbulent kinetic energy, mean and fluctuation concentrations around building. The main purpose is to evaluate the behavior of a contaminant gases released from different stack locations around buildings by various types of atmospheric stability: stable, neutral and unstable conditions.

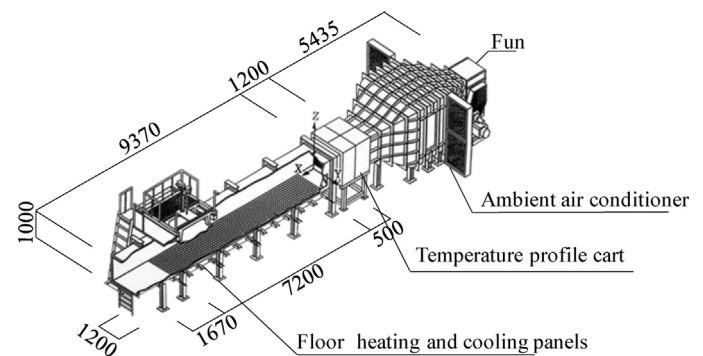


Fig. 1. Thermal stratified wind tunnel.

2. Experimental methodology

2.1. Experimental apparatus

Experiments were carried out in the thermally stratified boundary-layer wind tunnel at Tokyo Polytechnic University, Japan, under the following atmospheric stabilities: stable, neutral and unstable. The wind tunnel working section is $1.2\text{ m} \times 1.0\text{ m}$ and length is 16.0 m (Fig. 1). The wind tunnel consists of a fan, temperature profile cart, floor heating and cooling panels, and ambient air conditioner. Air temperature is controllable in the range $10\text{--}50\text{ }^{\circ}\text{C}$. The experiments were conducted with a wind velocity and air temperature at inlet of 1.3 m/s and $15\text{ }^{\circ}\text{C}$, respectively.

A simulated urban boundary layer was achieved using 26L-type roughness from aluminum plates, 9 mm high, and placed 200 mm apart on the wind tunnel floor (Fig. 2). This combination of roughness produced a simulated atmospheric boundary layer with a normal depth $\delta = 500\text{ mm}$ and a free stream wind speed $U_{\infty} = 1.3\text{ ms}^{-1}$. A widely used model to describe the vertical velocity distribution in the urban environment, in which the boundary layer is fully developed, can be described by a power law:

$$\frac{U_z}{U_H} = \left(\frac{Z}{H}\right)^n \quad (1)$$

where U_z is the mean velocity at elevation Z , U_H is the mean velocity at the building height, and n is the power exponent. We used the typical value $n = 1/4$ for an urban area. However, there is a range of other possible values for n . For details on typical values for different terrain types, refer to Snyder [22]. Fig. 3 shows the simulated turbulent boundary layer in the stratified wind tunnel under the three atmospheric stabilities (stable, neutral and unstable) at $X=0.0$. Fig. 4 shows typical temperature profiles in the vertical direction for a stable and unstable stratified boundary layer at $X=0.0$. Wind profiles show small variation up to 500 mm in Fig. 3, except for the region near the floor, where flow is highly turbulent because of the effect of roughness. The stratified flows show almost linear temperature profiles in the streamwise direction.

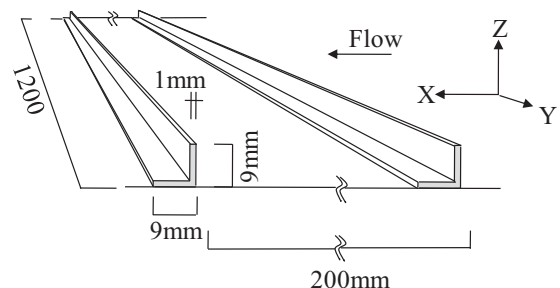


Fig. 2. Aluminum plates as roughness elements.

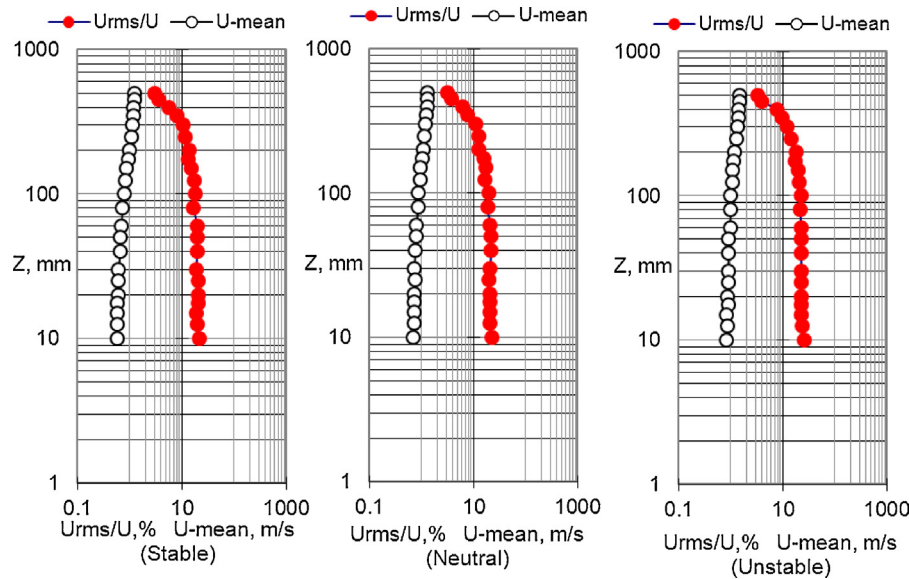


Fig. 3. Vertical distributions of mean velocity and turbulence intensity in the simulated boundary layer under thermal stability.

The three-dimensional isolated cubic building model has dimensions $100 \text{ mm} \times 100 \text{ mm} \times 100 \text{ mm}$. The model was constructed of steel. The simulated scale of the model in the urban boundary layer was 1:300. The model was located at $X/H = 0.0$. The stack was modeled by a steel cylinder with 7 mm height, with inner and outer diameters 4 and 6 mm, respectively. The stack was set at the right, center and left of the building roof. Fig. 5 shows a schematic of the building with stack locations.

2.2. Measurement instrumentation

A split fiber probe was adopted for wind velocity measurement, and a cold-wire thermometer used for temperature measurement. To precisely compensate for the contribution of temperature fluctuation to output voltages of the split film under the low wind velocity condition, it was necessary to obtain precise calibration data under the stable wind velocity and temperature conditions, thereby

improving calibration precision and efficiency. The split film and cold-wire were set adjacent at distance $\Delta Y = 2 \text{ mm}$, respectively, to measure fluctuating velocity and temperature simultaneously. The sampling frequency and low-pass filter cutoff frequency were set at 1000 Hz and 200 Hz, respectively; 60,000 data points were sampled in 60 s.

The stack gas used as a tracer gas in diffusion measurement was ethylene (C_2H_4), without buoyancy. Thus, buoyancy effects were neglected. A high-response hydrocarbon analyzer detector (flame ionization detector, FID) was used to measure C_2H_4 concentration. Each measurement was made for 90 s and produced an average value. The background was also recorded. The difference between the local and background reading was taken as the final concentration measurement. Concentration measurements are presented as the general non-dimensional concentration, K

$$K = \frac{C}{C_0} \quad (2)$$

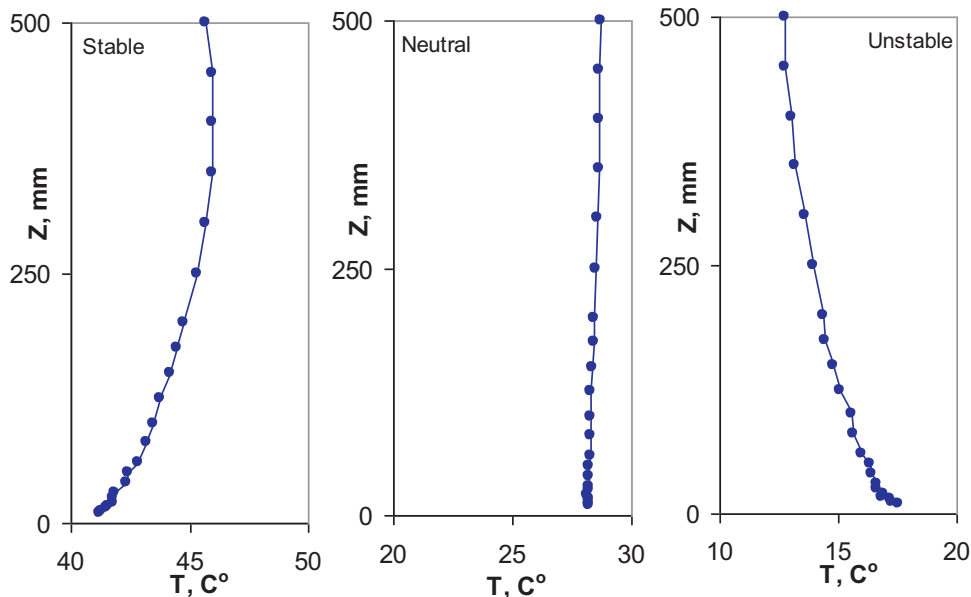


Fig. 4. Vertical profile of temperature in the simulated boundary layer.

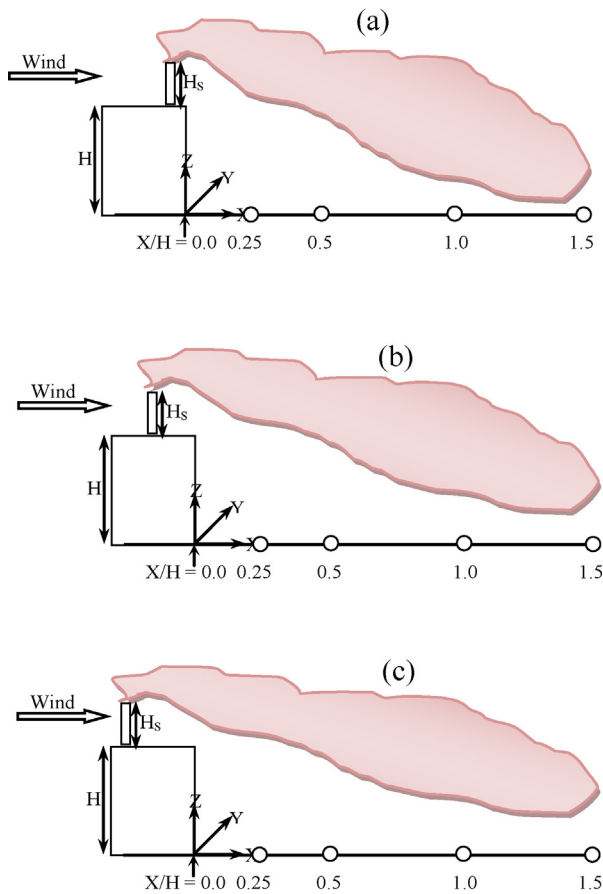


Fig. 5. Building and stack locations on: (a) right, (b) center & (c) left building roof.

$$C_0 = \frac{Q}{U_H H^2} \quad (3)$$

where C is measured concentration and C_0 the reference concentration, in which Q is the source volume flow rate and U_H the free stream velocity at building height H . Exhaust momentum from the stack was 50% of free stream velocity. Since the density of C_2H_4 gas is nearly that of air, the density of pollutant gas can be thought to have the same density at the height of the pollutant effluent in the boundary layer.

3. Simulation criteria

Simulation criteria for the wind tunnel experiments on atmospheric diffusion primarily include the following: (a) the geometry of the topography; (b) the ratio of pollutant release speed to that of the airflow, and atmospheric thermal stability; and (c) the surface boundary layer characteristics. Further, the three parameters representing the ratio of inertial properties to molecular properties are: (1) the Reynolds number Re ; (2) the Péclet number; and the Schmidt number. For more details on the similarity criteria, see Snyder [22,23].

3.1. Atmospheric thermal stability criteria

The three different thermal stratifications (stable, neutral and unstable) within the atmospheric boundary layer were created in the test section by controlling the wind flow temperature and wind

Table 1

Bulk Richardson number (Ri_B) and building and stack Reynolds numbers (Re_b & Re_s) in stratified thermal stability.

Atmosphere stability	Ri_B	Re_b	Re_s
Neutral	0.0	5.8×10^3	1.2×10^2
Stable	0.023	5.1×10^3	1.03×10^2
Unstable	-0.016	6.6×10^3	1.3×10^2

tunnel floor temperature. The atmospheric stability was evaluated with the Bulk Richardson number Ri_B

$$Ri_B = \frac{gz^2(dT/dz + \Gamma_{ad})}{TU^2} \approx \frac{gH\Delta\theta}{TU^2} \quad (4)$$

where H is the reference height, T is temperature at H , and $\Delta\theta$ is the temperature between the reference height and the ground level. The resulting Ri_B in atmospheric stability is shown in Table 1. The value of Ri_B in the current experiment was sufficient to create the thermally stratified boundary layer.

3.2. Flow and dispersion criteria

Wind tunnel modeling criteria for accurate flow and dispersion around the building were studied in detail by Snyder [22]. The following criteria are generally suggested for gaseous exhausts without buoyancy under neutral stratified conditions in wind tunnel dispersion studies:

- Geometric similarity.
- Similarity of wind flow in wind tunnel to the atmospheric surface layer.
- Building Reynolds number ($Re_H = U_H H / \nu$) should be greater than 1.1×10^4 .
- Stack Reynolds number ($Re_s = U_s d_s / \nu$) should be greater than 2×10^3 .
- Equivalent stack momentum ratio ($M = U_s / U_H$).

where ν is the kinematic viscosity of air, U_H is the wind velocity at building height, d_s is outlet diameter of the stack, and U_s is exit velocity of the stack. Furthermore, Castro and Robins [24] and Snyder [25] suggested that Re_b should be greater than 4×10^3 . Resulting Reynolds numbers based on Re_b and Re_s in thermal stability are shown in Table 1. On the other hand, Hoult and Weil [26], White and Stein [2] and Lape [27] that showed Re_s as low as 300, 533 and 700 were adequate for a buoyant plume, respectively. In wind tunnel experiments for pollutant dispersion it is very important to maintain turbulent flow around the building and stack. Resulting Reynolds numbers based on Re_b and Re_s in thermal stability are shown in Table 1. Despite the low value of Re_s , a trip was not used in the stack. The low exhaust momentum caused the plume to be trapped in a separate flow region and consequently, the effect of laminar exhaust flow should be negligible. Furthermore, Saathoff et al. [8] suggested that “it is generally not possible to satisfy the stack Reynolds number for small diameter stacks and it is also difficult for the flow to travel for such stacks.” Although the stack Reynolds number was somewhat lower than 2000, this may have had minimal effect on measurement results, as discussed in Hajra et al. [10].

4. Results and discussion

To increase understanding of contaminations transportation and dispersion emitted from short stacks on the building roof under changes of stack locations and atmospheric stability, it is necessary to know the flow field characteristics. Different changes of stack locations and atmospheric stability produce different flow patterns

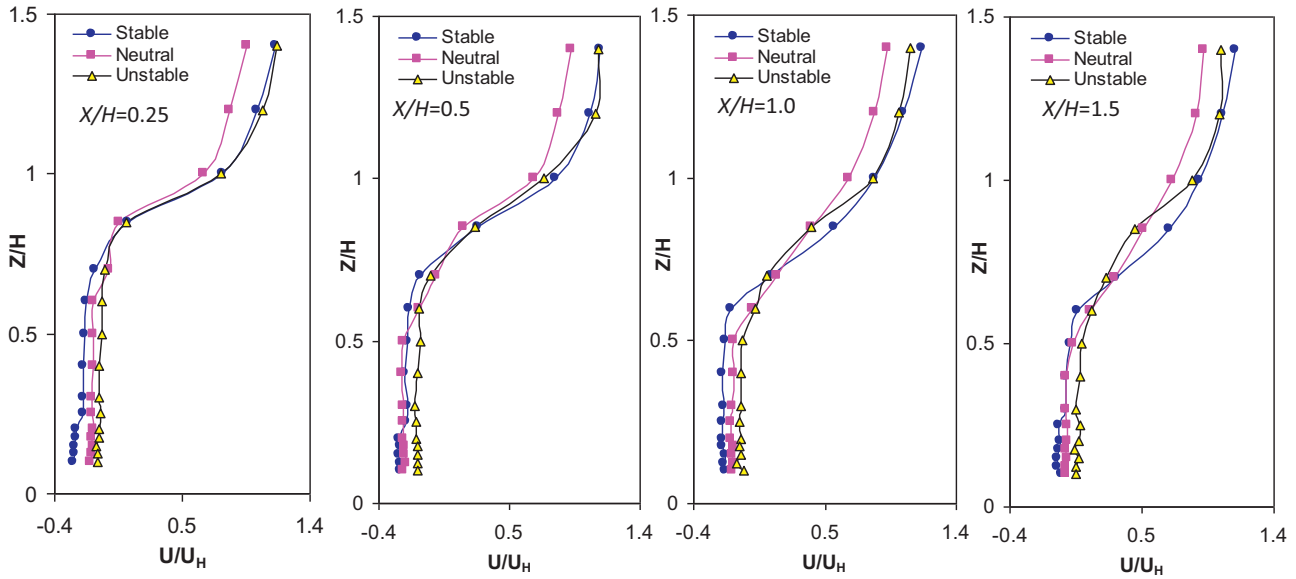


Fig. 6. Mean velocity component in the longitudinal direction, U/U_H .

around buildings, this cause a change in the characteristics of contaminate dispersion. Measurements were made at four different downwind locations along the wind tunnel centerline, $X/H = 0.25, 0.5, 1.0$ and 1.5 (Fig. 5). Flow and pollutant measurements in the turbulent boundary layer were made at various heights, beginning at 10 mm above the bottom of the wind tunnel.

4.1. Flow simulation

4.1.1. Mean velocity

Mean velocity components were normalized by the obtained mean velocity U_H , which is the approaching wind velocity at building height H in the thermal atmospheric boundary layer. Figs. 6 and 7 show the vertical profiles of the mean velocity in the longitudinal and vertical directions (U, W) around the building, under the stable, neutral and unstable atmospheric conditions.

In particular, when the thermal stability was strong, the wind velocity in the lower part around the building was very low,

dropping to 1 or 2% of the wind tunnel velocity. The longitudinal velocity increased rapidly upward, starting at $0.6 > Z/H < 1$ and gradually reaching $Z/H > 1$. The velocity increase at roof level was from flow shear. The longitudinal velocity appeared very thick around the building at $X/H = 0.25, 0.5$, and 1 . This is from the substantial effect of turbulent mixing, which is created by the flow distortion around the building. This creates a small separation in this region. A thin vertical profile was generated at $X/H = 1.15$. The unstable conditions increased the longitudinal velocity at the lower height of $Z/H = 0.6$. The stable profiles showed decreased longitudinal velocity at this lower height. This is caused by increasing turbulence, which augments momentum transfer from higher to lower levels. The reattachment length of the separated flows under stable conditions at $X/H = 0.25$ was longer than that at the other locations.

The vertical velocity clearly indicates an upward motion for the flow. The vertical velocity appeared very thick at $X/H = 0.5, 1$ and 1.5 . This is from the strong effect of turbulent mixing, created by

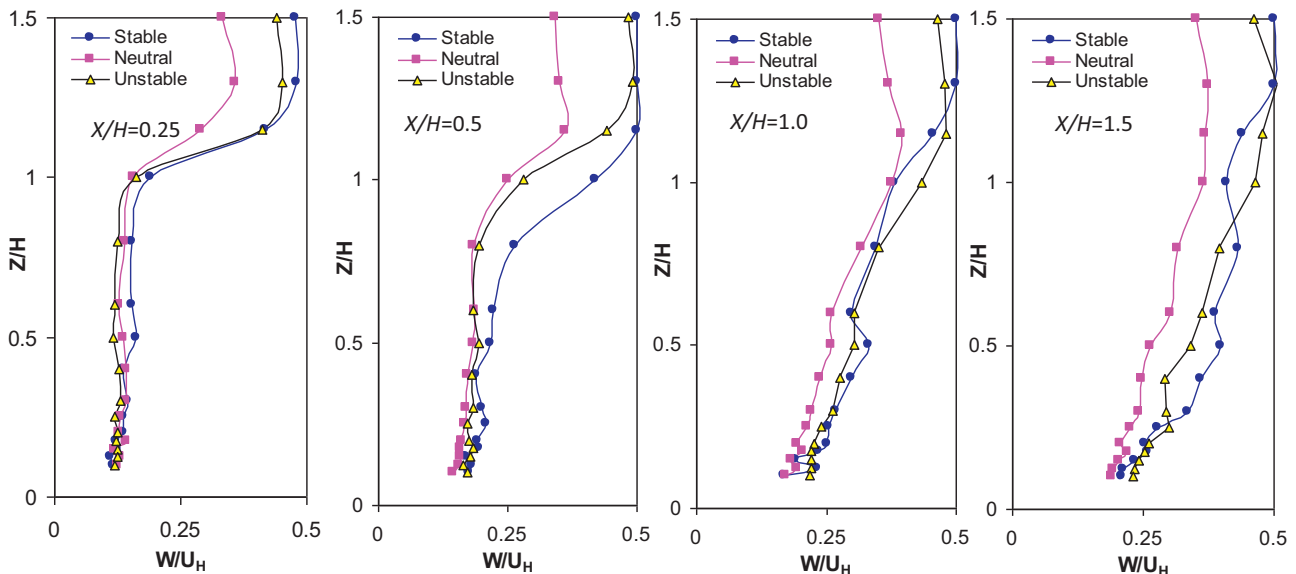


Fig. 7. Mean velocity component in the vertical direction, W/U_H .

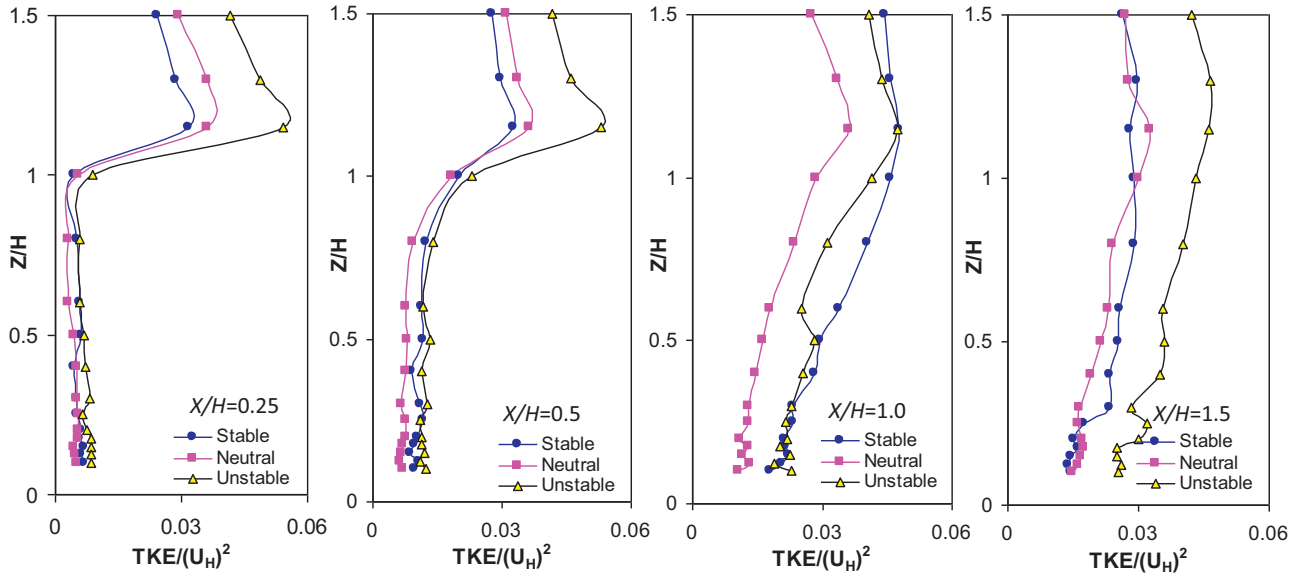


Fig. 8. Turbulent kinetic energy, $TKE/(U_H)^2$.

flow distortion in the downwind direction. The vertical velocity was increased and directed upward at $X/H=0.5$, 1, and 1.5, because of the steep gradient of the vertical turbulence intensity. The change in vertical velocity at $X/H=0.25$ was small. This is because of less intense vertical turbulence. The vertical velocity increased under stable conditions because of increased turbulence, which augments momentum transfer from higher to lower levels. The mean velocity in the vertical direction was smaller for the unstable case than for the neutral and stable cases. In general, the mean velocity in the vertical direction for stable conditions was higher than that in unstable conditions (Snyder [22]; Panofsky et al. [28]; Tennekes and Lumley [29]), because: (a) the heat flux is positive due to upward heat transfer; (b) the production of turbulent kinetic energy is increased; (c) the buoyant production is positive and (iv) the bulk Richardson number is positive. In addition to the turbulence being smaller in stable than in unstable conditions, the most important influence observed was in the vertical direction (Ogawa et al. [30]; Zegadi et al. [31]).

This result shows that when the atmospheric stratification is stable, the vertical velocity increases around the building. When this stratification is unstable, the vertical velocity decreases and the longitudinal velocity increases around the building.

4.1.2. Turbulent kinetic energy

Fig. 8 shows the vertical distributions of the turbulent kinetic energy TKE around the building. The changes of atmospheric stability on turbulent energy are very significant and strongly affect the velocity field and concentration distributions. Greater turbulent energy always generates a larger turbulent spread of the plume; also the size of the region of flow influenced by the building affects the plume spread. It is seen that the high turbulent energy is even stronger close to the roof surface. This behavior is mainly related to intense temperature gradients in this region, which create strong buoyancy effects. Variations of stratification were strong far from the stack at $X/H=1.0$ and 1.5. Turbulent energy under unstable conditions was greater than under neutral and stable conditions.

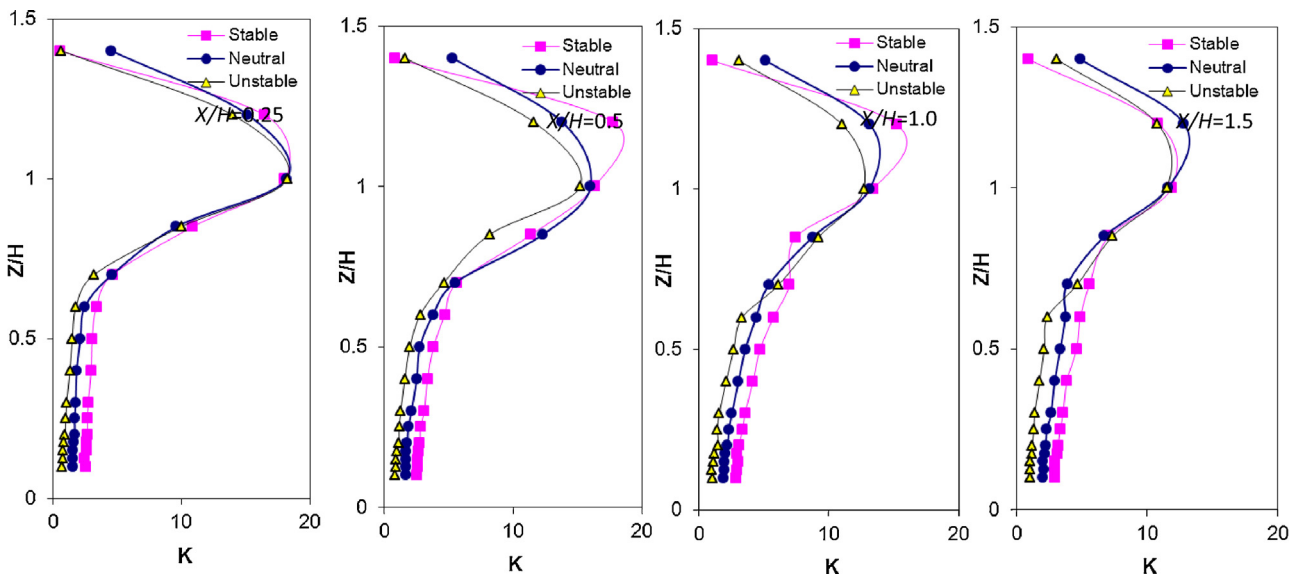


Fig. 9. Vertical profiles of mean concentration, K for stack located on the right building roof.

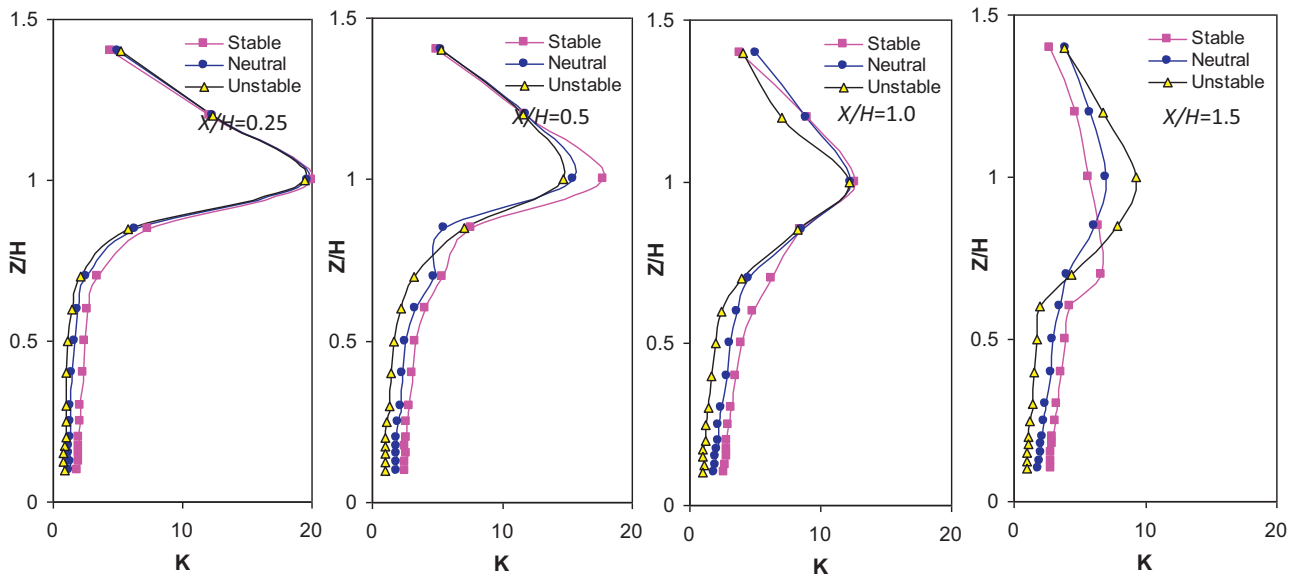


Fig. 10. Vertical profiles of mean concentration, K for stack located on the center building roof.

Turbulent energy in unstable conditions was much higher at $X/H = 1$ and 1.5, because of a large velocity gradient. The lowest value of the turbulent energy was in neutral conditions. Minimum turbulent energy was found in the lower portion around the building. Peaks of turbulent energy were observed in its upper portion, near the stack. This energy increased upwards gradually, and reached a maximum near the roof surface. This is because of shear production by the turbulent kinetic energy. The effect of atmospheric stability on turbulent energy was much greater with increasing distance from the building. This result shows that as turbulent energy increases, a smaller portion of the plume is carried upwind because of a reduction in the size of the reverse flow region.

4.2. Dispersion simulation

Contaminant dispersion is also significantly affected by variability of stack locations and atmospheric stability. Concentration measurements were conducted in the wind tunnel experiment through the atmospheric stability around the building. The stack

model was located in the right, center and left of the building roof. Figs. 9–11 show the vertical distributions of concentration measured in the atmospheric stability for different stack location on the building.

For different atmospheric stratifications, dispersion distributions varied significantly downstream of the buildings. Near the buildings, however, dispersion was dominated by mechanical turbulence generated by the building itself. In particular, when effluents emerge from the vertical stack at low momentum or low mean vertical velocity, and horizontal flow around the stack is sufficiently strong, the effluent plume may be dragged down in the low-pressure region in the wake of the stack. This phenomenon is referred to as stack downwash (Ogawa et al. [32]; Arya [33]). The vertical distributions of concentration clearly show downwash around the building. They are highly skewed in the upward direction; the negative tail is essentially in the downward direction for $Z/H < 1$. In spite of the highly non-Gaussian character of the concentration distributions close to the stack, they tend to increase Gaussian at a short distance downwind the building. Moreover, it

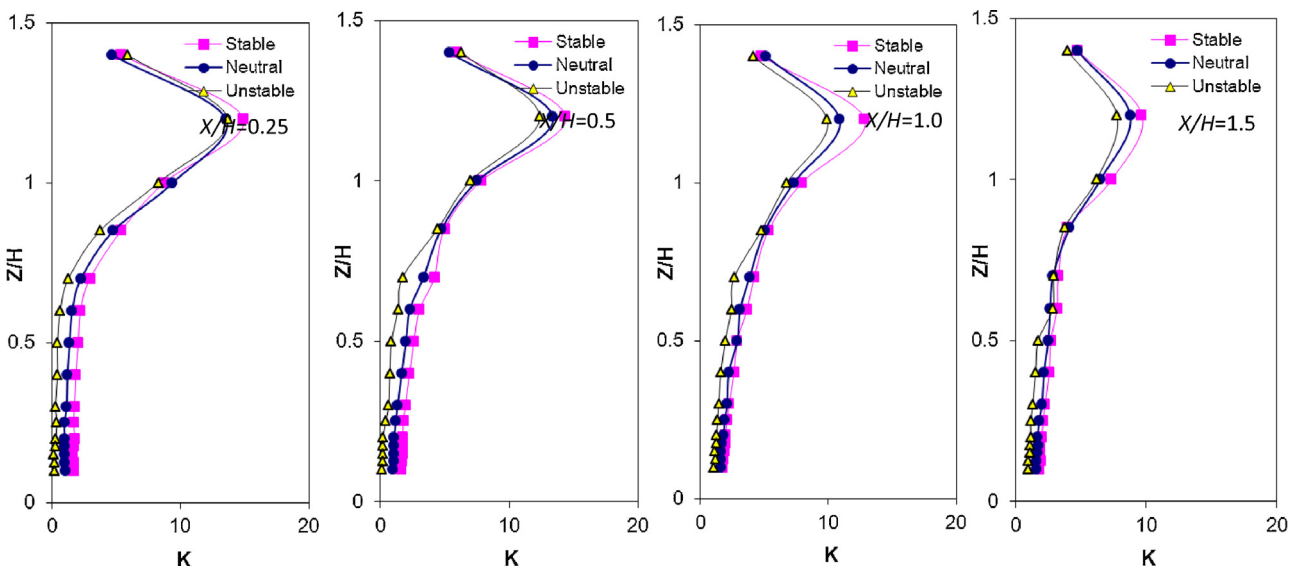


Fig. 11. Vertical profiles of mean concentration, K for stack located on the left building roof.

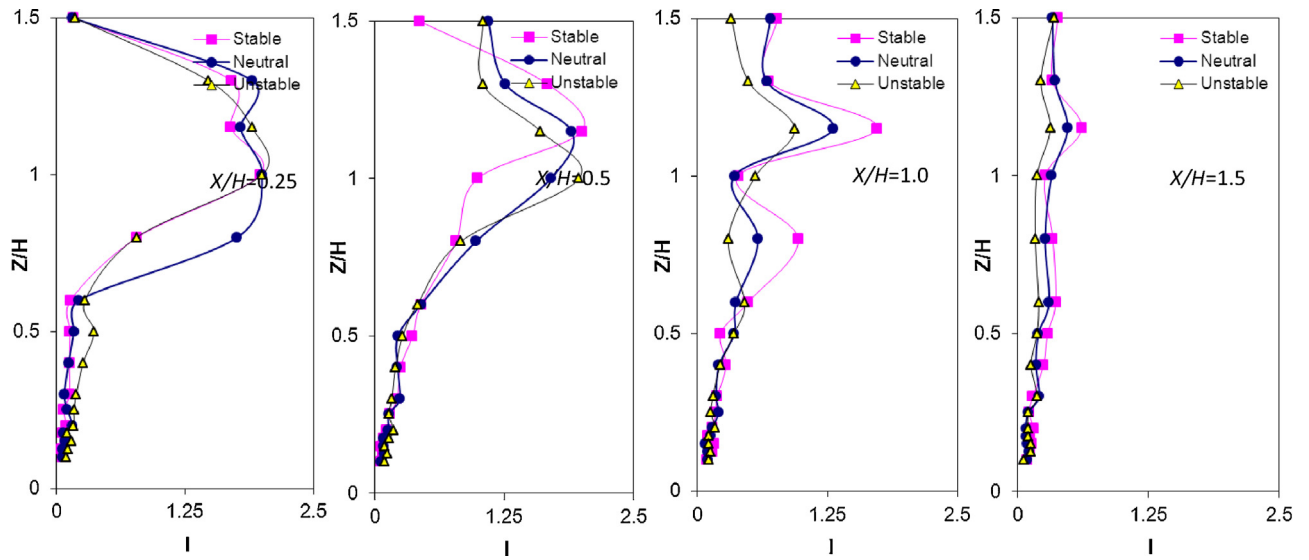


Fig. 12. Vertical profiles of concentration fluctuation, I for stack located on the right building roof.

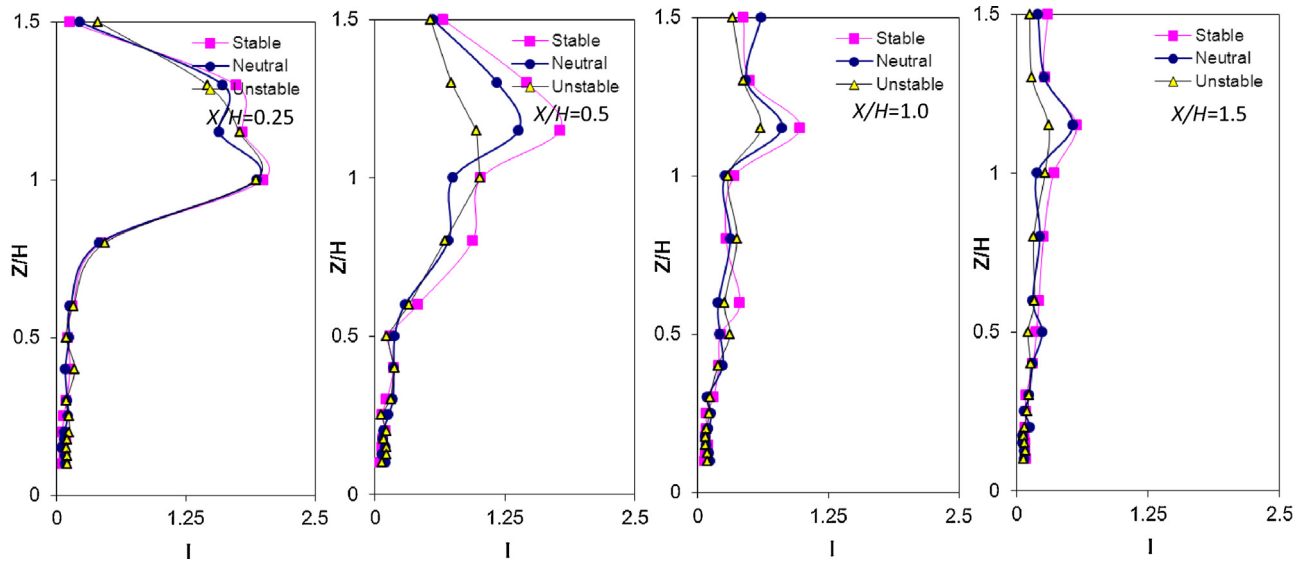


Fig. 13. Vertical profiles of concentration fluctuation, I for stack located on the center building roof.

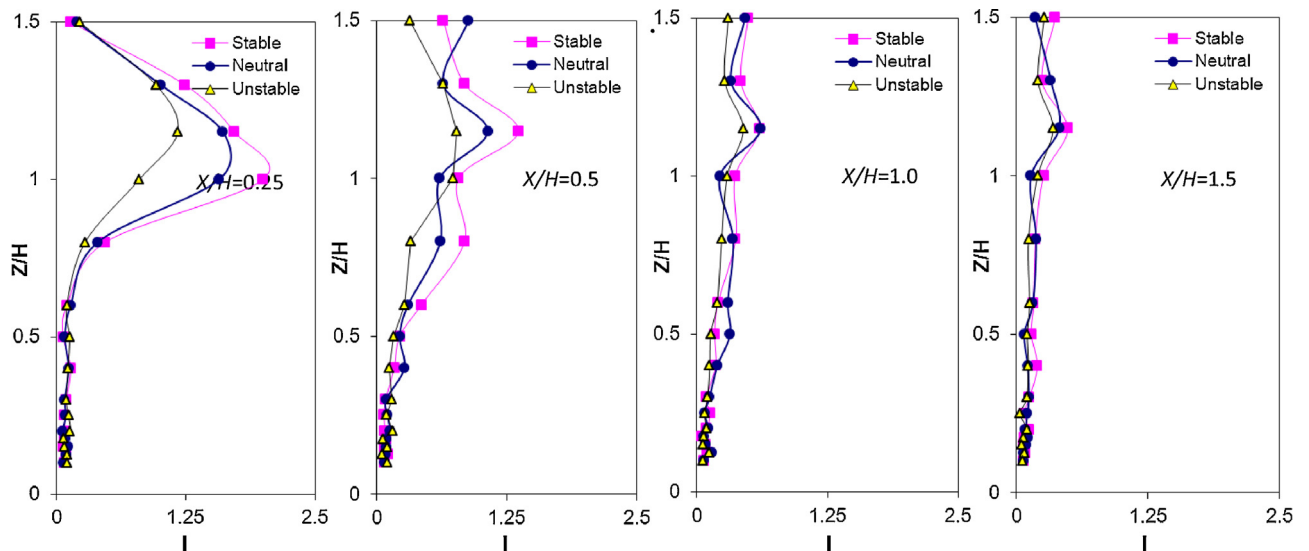


Fig. 14. Vertical profiles of concentration fluctuation, I for stack located on the left building roof.

is clear that the centroid of these distributions descends slightly with downwind distance. This result appears consistent with the observations of Snyder and Lawson [34].

The concentration distributions gradually increased upward starting at $0.75 < Z/H < 1$, reaching a maximum at $Z/H = 1.0$, and gradually decreased at $Z/H > 1.0$. Variations of the concentration distributions were weak far from the stack at $X/H = 1.0$ and 1.5 . This is due to the weak of effluent momentum of contaminations. The lowest value of concentration was for the lift stack on the building. While, the highest value of concentration was for the right stack on the building. Minimum concentration was found in the lower portion around buildings. Peaks of concentration were observed in its upper portion, near the building roof. Concentration distributions increased for the right stack and decreased for the lift stack on the building.

The concentration distributions increased under stable conditions and decreased under unstable conditions around the building. This result shows that air changes around the building were affected mainly by turbulent mixing between around the building and the free stream regions. Peak concentrations of atmospheric stability were observed at $Z/H = 1$, where the effluent was emitted near the separation–reattachment region, thus causing the downwash. This is because emission momentum from the stack was less than the free stream velocity. The maximum concentration was at $X/H = 0.25$, near the stack emission on the roof surface. The minimum concentration on the roof surface was at $X/H = 1.5$, far from the stack emission. Therefore, the dispersion concentration was high near the stack emission and decreased with distance from the stack. On the other hand, minimum concentration around the building was at $X/H = 0.25$ under unstable conditions. This is from the higher vertical turbulent velocity. Contaminant concentration around the building at $Z/H < 0.75$ under unstable conditions was lower than that under neutral and stable conditions. This is because the vertical velocity was lower in this region. However, the concentration over the building roof ($Z/H > 1$) at $X/H = 1.5$ (far from the stack) under unstable conditions was higher than that under neutral and stable conditions, consistent with Ogawa et al. [32]. This is because the turbulent diffusion of concentration is reduced in this region. Contaminate concentration around the building at $Z/H > 1$ changed slightly with atmospheric stability, except at $X/H = 1.5$.

This result shows that the change of atmospheric stability conditions around the building is to increase the concentration under stable conditions, and decrease it under unstable conditions. The concentration distributions increased for stack located at the right and decreased stack located at the right on the building.

The vertical distributions of concentration fluctuation are shown in Figs. 12–14. The variations of Concentration fluctuation were strong for stack located at the right on the building roof. The changes of stack locations and atmospheric stability is clear at $Z/H > 0.5$. The versions of concentration fluctuations for the center stack appear similar to the lift stack on the building. The concentration fluctuations show a pattern similar to the mean concentration in the lower part around the building. The changes of stack locations and stability on the concentration fluctuations was slight at $Z/H < 0.75$, near the stack, but increased rapidly through $Z/H = 1.0$ and generally decreased at $Z/H > 1.0$. The concentration fluctuations for the right stack were greater than those for the center and lift stacks of the building. This is due to the strong of effluent momentum of the right stack. Concentration fluctuations under stable conditions at $Z/H > 1$ were greater than those under unstable conditions. This is a result of the strong influence of the turbulent energy in the region. Smaller concentration fluctuations for different stack locations were observed under unstable conditions in the upper portion around the building. Fluctuations in its lower portion were close to zero, because the plume spread was so narrow that only very small eddies could be generated within it. The

concentration fluctuations were less than 1 at $Z/H < 1$, but reached a maximum of 2 near the roof surface at $X/H = 0.25$. The fluctuations slowly decreased at distances far downwind of the stack.

5. Conclusion

The effect of variability in stack locations and atmospheric stability on the flow and dispersion of building exhaust emissions in urban environment was studied experimentally using a thermal boundary layer wind tunnel. It was shown that both stack locations and atmospheric stability had a strong influence on the flow and dispersion behavior.

The results show that the stable condition decreased the longitudinal velocity in the lower part around the building and increased the vertical velocity around the building. The unstable condition decreased the vertical velocity and increased the longitudinal velocity around the building. The high turbulent energy is even stronger close to the roof surface. Turbulent energy under unstable conditions was greater than under neutral and stable conditions. Minimum turbulent energy was found in the lower portion around the building. Peaks of turbulent energy were observed in its upper portion, near the stack. The lowest value of the turbulent energy was in neutral condition.

The vertical distributions of the concentration clearly showed downwash around the building. The concentration distributions increased for the right stack and decreased for the lift stack on the building. The lowest value of the concentration was for the lift stack on the building. The highest value of the concentration was for the right stack on the building. The concentration distributions increased under stable conditions and decreased under unstable conditions around the building. The concentration fluctuations under stable conditions at $Z/H > 1$ were greater than those under unstable conditions. Smaller concentration fluctuations were observed under unstable conditions in the upper portion around the building. The concentration fluctuations for the right stack were greater than those for the center and lift stacks of the building.

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